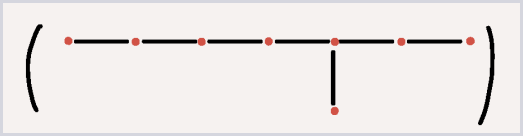
Lecture 1 // Aug 25 // Overview

Theorem 1 (Freedman)
Let Q be a unimodular symmetric bilinear form. Then there exists a unique simply connected topological 4-manifold X^4 such that $Q_X = Q$.

1. If Q is even, there exists a unique smooth X^4 up to homeomorphism.
2. If Q is odd, there exist at most two smooth X^4 up to homeomorphism, of which at most one is smoothable.

Theorem 2 (Donaldson)
If X^4 is a simply connected, smooth, closed 4-manifold and Q_X is definite, then Q_X is equivalent to $n\langle+1\rangle$ or $n\langle-1\rangle$.

Example 3
E_8 matrix:



$$\text{adj} = \begin{pmatrix} -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & -2 \end{pmatrix}$$

Check: no change of basis takes this to $\pm I_8$.

$\Rightarrow E_8$ is realized by a non-smoothable topological X^4.
$\Rightarrow$ the same holds for nE_8.

Over the next few weeks, we'll discuss the proof of Donaldson's theorem. After, we'll discuss Instanton Floer Homology (IFH) and some applications to three-manifolds, in particular, the homology cobordism group. The rest of this lecture will be to lay the groundwork for this.

3-Manifolds

Definition 4 (Homology Cobordism Group)

$$\Theta^3_{\mathbb{Z}} = \{\text{all } \mathbb{Z}\text{HS}^3\text{'s}\}/\text{homology cobordism.}$$

$Y_1 \sim Y_2$ are homology cobordant if there exists W^4, a homology $S^3 \times I$, such that $\partial W^4 \cong -Y_1 \sqcup Y_2$.

- The topological version is trivial.

Theorem 5 (Freedman)
If Y is a $\mathbb{Z}\text{HS}^3$, then $Y = \partial W$ for some $\mathbb{Z}\text{HB}^4$. In fact, $\pi_1 W = 1$.

Proof. Given a unimodular symmetric bilinear pairing Q, we can construct a smooth X^4 with boundary and intersection form Q by attaching 2-handles to B^4 according to Q:

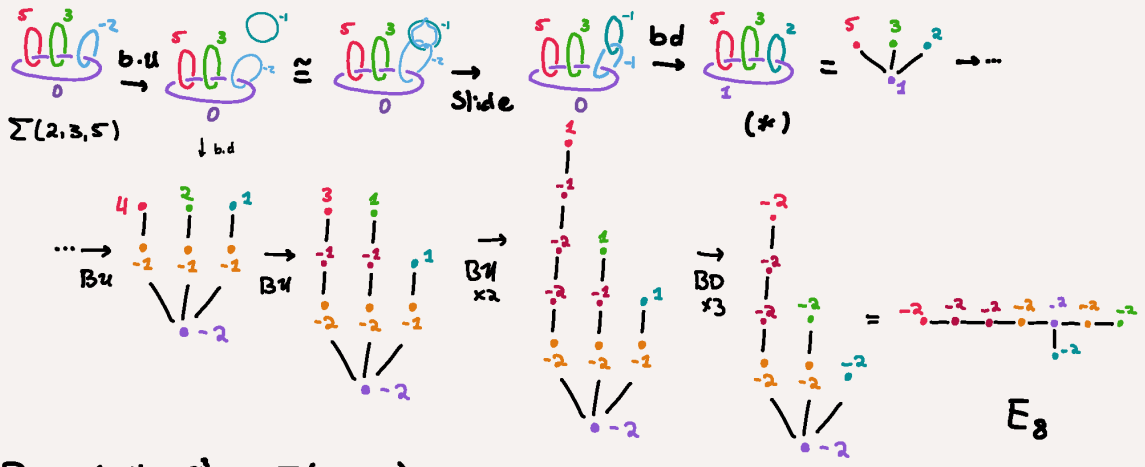


E_8:

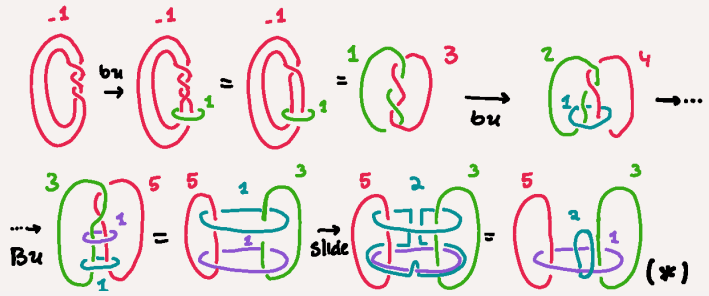
For E_8, $\partial E_8 \cong$ Poincaré homology sphere $\cong \Sigma(2,3,5)$.
By Donaldson, $\Sigma(2,3,5)$ does not bound any smooth $\mathbb{Z}\text{HB}^4$

$$\Rightarrow [\Sigma(2,3,5)] \neq 1 \in \Theta^3_{\mathbb{Z}}.$$

Proof of boundary-diffeomorphism $\Sigma(2,3,5) \cong \partial E_8$:



Poincaré Hx $S^3 \cong \Sigma(2,3,5)$



□

Remark 6

There are many $\mathbb{Z}\mathrm{HS}^3$'s.

Example 7

Brieskorn spheres: p, q, r pairwise coprime,

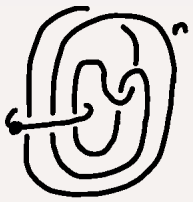
$$\Sigma(p,q,r) := S^5_\epsilon \cap \{x^p + y^q + z^r = 0\} \subseteq \mathbb{C}^6.$$

Example 8

$S^3_{1/n}(k)$ for all k.

Example 9

Mazur manifolds: 1-handle ∪ 2-handle such that the algebraic linking is ± 1, but the geometric linking is $\neq \pm 1$.



Question 10. *How large is* $\Theta^3_{\mathbb{Z}}$*?*

Theorem 11 (Furuta, FS)
$\{\Sigma(2,3,6n-1)\}$ is a $\mathbb{Z}^\infty$ subgroup of $\Theta^3_{\mathbb{Z}}$.

Theorem 12 (Daemi–Nozaki–Sato–Taniguchi)
Sometimes $\{S^3_{1/n}(k)\}$ are linearly independent.

This relies on a specific condition on IFH_*.

Note 13
$\Sigma(2,3,6n-1) = S^3_{1/n}(T_{2,3})$.

From concordance to homology cobordism:

$$\Sigma_2 : \mathcal{C} \longrightarrow \Theta^3_{\mathbb{Z}/2}, \qquad k \longmapsto \Sigma_2(k) = \text{double branched cover over } k,$$

is a homomorphism.

Question 14. *Is there a distinction between bounding a* $\mathbb{Z}\mathrm{HB}^4$ *and a* $\pi_1 = 0$ $\mathbb{Z}\mathrm{HB}^4$*? I.e., is there a* $\pi_1 = 1$ *homology cobordism from* $\Sigma(2,3,5)$ *to itself?*

Theorem 15 (Taubes)
No.

Theorem 16 (Daemi–Nozaki–Sato–Taniguchi)
Often, no. Cutting open the cobordism, we get a simply connected $\mathbb{Z}\mathrm{HB}^4$ bounding $Y\#-Y$.